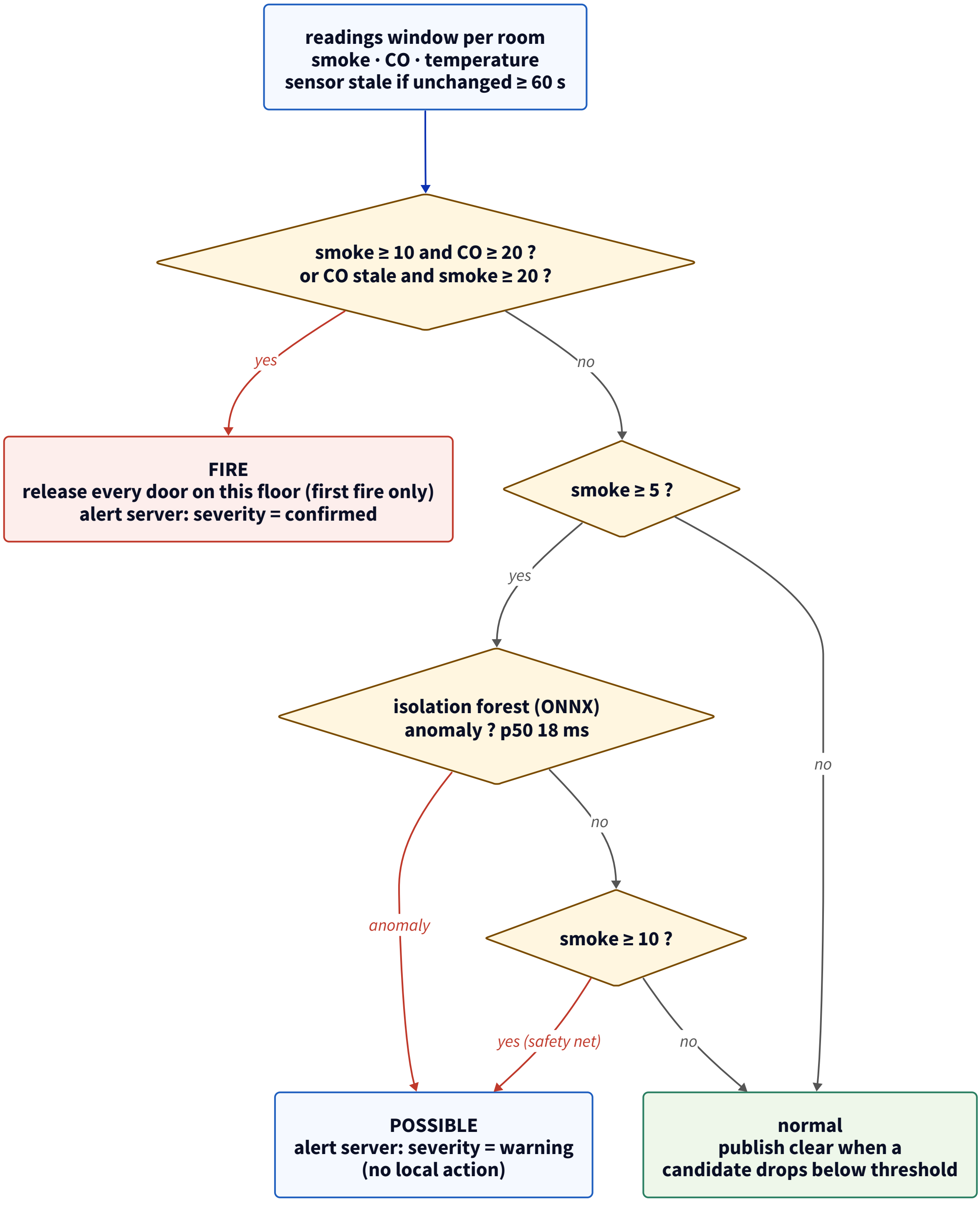

readings window per room
smoke · CO · temperature
sensor stale if unchanged ≥ 60 s
smoke ≥ 10 and CO ≥ 20 ?
or CO stale and smoke ≥ 20 ?
yes
no
FIRE
release every door on this floor (first fire only)
alert server: severity = confirmed
smoke ≥ 5 ?
yes
no
isolation forest (ONNX)
anomaly ? p50 18 ms
anomaly
no
smoke ≥ 10 ?
yes (safety net)
no
POSSIBLE
alert server: severity = warning
(no local action)
normal
publish clear when a
candidate drops below threshold